ft-de: functional tensor-train density estimation and deconvolution with Chebyshev cores

Notes on the `ft-de` repository

September 2026

Abstract

These notes describe the estimator implemented in `ft-de`. It estimates a probability density on $\mathbb{R}^d$ from samples, optionally when every sample is corrupted by additive noise with a known distribution. The density is the normalized square of a *functional* tensor train whose cores are matrix-valued Chebyshev expansions (the `ChebTT` objects of chebdGE). The Chebyshev coefficients are fitted by stochastic maximum likelihood. In the noisy case the likelihood is that of the *convolved* model $U * p$, where the noise density U is itself a Chebyshev tensor train. We show that this convolved likelihood, its normalization, all marginals, and the smoothness functionals used for regularization are evaluated exactly (to rounding) by small tensor-network contractions, without Monte Carlo averaging over the noise. The note then covers the regularizers and the data-only model-selection rules the method needs because deconvolution is ill-posed, and reports qualitative and quantitative experiments: the Ginzburg–Landau benchmark, a continuous "swirl–pretzel" benchmark introduced to probe ridge fragmentation, sample-size convergence, and a simulated cryo-EM latent space.

Contents

1 Setting and overview **2**

2 The model: squared Chebyshev tensor trains **3**
2.1 Exact integrals . . . 3

3 Fitting without noise **4**
3.1 Objective and optimizer coordinates . . . 4
3.2 Initialization by chebdGE cross approximation . . . 4
3.3 Capacity continuation . . . 5

4 Deconvolution: the exact convolved likelihood **5**
4.1 Model and ill-posedness . . . 5
4.2 Tensor-train noise kernels . . . 5
4.3 Exact evaluation . . . 6

5 Regularization **7**
5.1 Amplitude Sobolev seminorms . . . 7
5.2 Marginal Fisher-visibility filter . . . 7
5.3 Pilot marginals . . . 8

6 Model selection from noisy observations **8**

7 The two recipes **8**

8 Experiments **9**
8.1 Ginzburg–Landau 9
8.2 Sample-size convergence 9
8.3 A continuous benchmark: the swirl–pretzel tube 13
8.4 Cryo-EM latent space 17

9 Discussion and open problems **19**

A Code map and reproduction **23**

1 Setting and overview

Let $X_1, \dots, X_N$ be i.i.d. with an unknown density p on $\mathbb{R}^d$. In the *noisy* problem we observe instead

$$Y_i = X_i + \varepsilon_i, \qquad \varepsilon_i \sim U \text{ i.i.d., independent of } X_i, \tag{1}$$

with the noise density U known, so that Y_i has density $q = U * p$. The target is always the latent density p, not q. Both problems are motivated by cryo-EM heterogeneity analysis: a latent embedding of every particle image is estimated with a known (posterior) uncertainty [7], and one wants the density of conformations rather than of the noisy embeddings.

The estimator in one paragraph. Fix a box $\Omega = \prod_k [a_k, b_k]$, TT ranks $r_1, \dots, r_{d-1}$ and n Chebyshev modes per axis. Every coefficient tensor $\mathcal{A} = (A_1, \dots, A_d)$ defines an amplitude $g_{\mathcal{A}}(x) = G_1(x_1) \cdots G_d(x_d)$, where each G_k is a matrix-valued polynomial, and a density $p_{\mathcal{A}} = g_{\mathcal{A}}^2 / Z(\mathcal{A})$ on Ω. We minimize the empirical negative log-likelihood of $p_{\mathcal{A}}$ (noise-free) or of $U * p_{\mathcal{A}}$ (noisy). Optional gauge-invariant smoothness penalties are added. The optimizer is Adam, run in L^2-orthonormalized coefficient coordinates, from a chebdGE cross approximation of $\sqrt{\text{KDE}}$. Hyperparameters (rank, degree, regularization strength) are chosen only from held-out observations, noisy ones in the noisy case, by a paired one-standard-error rule.

Relation to TTDE. The closest antecedent is TTDE [13]. Its table-data model (`PAsTTSqrOpt`, used in the earlier cryo-EM notebooks of this project) is

$$p(x) \propto \sum_{c=1}^{C} \langle \mathcal{A}_c,\, b(x_{\pi_c(1)}) \otimes \cdots \otimes b(x_{\pi_c(d)}) \rangle^2, \tag{2}$$

a mixture over coordinate permutations π_c of squared TT contractions against a B-spline feature vector b, trained by Riemannian Adam. (Other TT density estimators build the TT by sketching rather than likelihood optimization [11, 16].) The present estimator keeps the key structural idea, a *squared* TT whose normalization and marginals are exact contractions. It differs in five respects, which organize these notes.

1. **Basis and representation.** The cores are Chebyshev coefficient tensors, i.e. exactly chebdGE's `ChebTT` with coefficient cores. Every integral we need reduces to closed-form Chebyshev Gram matrices and moments (§2). There is a single TT, not a permutation mixture.
2. **Initialization** by chebdGE's cross approximation (TT-qTensGE) of $\sqrt{\text{KDE}}$ (§3).

3. **Exact deconvolution likelihood.** If U is a Chebyshev TT, $\log(U * p_{\mathcal{A}})(y)$ is evaluated exactly by a transfer-matrix contraction with precomputed polynomial "moment cores" (§4). The earlier TTDE cryo-EM pipeline replaced $q(y)$ by a Monte Carlo average $M^{-1}\sum_m p(y-\varepsilon_m)$ over noise draws, which biases $\log q$ (Jensen) and adds gradient noise.
4. **Regularization** by gauge- and scale-invariant functionals of the *represented amplitude*: integrated derivatives and a marginal Fisher-information filter (§5).
5. **Selection from noisy data only** (§6). Clean data and the true density enter only in post-hoc diagnostics.

Notation. T_i is the Chebyshev polynomial of the first kind; $\tau_k(x) = (2x - a_k - b_k)/(b_k - a_k)$ maps $[a_k, b_k]$ to $[-1, 1]$. Cores have shape $A_k \in \mathbb{R}^{r_{k-1}\times r_k \times n_k}$ with $r_0 = r_d = 1$, and $A_k^{(i)} \in \mathbb{R}^{r_{k-1}\times r_k}$ is the i-th slice. We write $r = \max_k r_k$ and $n = \max_k n_k$. The relative error of an estimate $\hat{p}$ is always the unsquared full-dimensional Lebesgue quantity $\operatorname{rel} L^2(\hat{p}) = \|\hat{p} - p\|_{L^2(\mathbb{R}^d)}/\|p\|_{L^2(\mathbb{R}^d)}$.

2 The model: squared Chebyshev tensor trains

Definition 1 (Chebyshev functional TT)**.** A *Chebyshev TT* on Ω is a function

$$g(x) = G_1(x_1)G_2(x_2)\cdots G_d(x_d), \qquad G_k(x_k) = \sum_{i=0}^{n_k-1} A_k^{(i)}\, T_i(\tau_k(x_k)) \in \mathbb{R}^{r_{k-1}\times r_k}. \tag{3}$$

This is the continuous (functional) version [1, 8] of the TT format [15] in a Chebyshev basis [21]. It is chebdGE's `ChebTT` with `core_representation="coeff_tensor"`. The density model is

$$p_{\mathcal{A}}(x) = \frac{g_{\mathcal{A}}(x)^2}{Z(\mathcal{A})}\,\mathbf{1}_\Omega(x), \qquad Z(\mathcal{A}) = \int_\Omega g_{\mathcal{A}}^2. \tag{4}$$

Squaring enforces $p_{\mathcal{A}} \geq 0$ for every real $\mathcal{A}$, as in TTDE and in the squared-TT samplers of [4, 5]. The map $\mathcal{A} \mapsto p_{\mathcal{A}}$ is invariant under (i) the TT gauge $G_k \mapsto G_k S$, $G_{k+1} \mapsto S^{-1}G_{k+1}$, (ii) scaling of any core, and (iii) $g \mapsto -g$. All functionals below are chosen to respect these invariances. This is why penalties are expressed through the represented amplitude, never through entrywise core norms.

2.1 Exact integrals

All integrals reduce to the one-dimensional Chebyshev moments and Gram matrices

$$\mu_s = \int_{-1}^{1} T_s = \begin{cases} \frac{2}{1-s^2} & s \text{ even} \\ 0 & s \text{ odd} \end{cases}, \qquad M_k[i,j] = \int_{a_k}^{b_k} T_i T_j \circ \tau_k = \frac{b_k - a_k}{4}\big(\mu_{i+j} + \mu_{|i-j|}\big), \tag{5}$$

using $T_iT_j = \frac{1}{2}(T_{i+j} + T_{|i-j|})$. Define the *Gram transfer map* of core k acting on $S \in \mathbb{R}^{r_{k-1}\times r_{k-1}}$,

$$\mathcal{E}_k(S) = \sum_{i,j} M_k[i,j]\,(A_k^{(i)})^{\mathsf{T}} S\, A_k^{(j)} \in \mathbb{R}^{r_k\times r_k}. \tag{6}$$

Then $Z(\mathcal{A}) = \mathcal{E}_d \circ \cdots \circ \mathcal{E}_1(1)$, at cost $O(d(r^3 n + r^2 n^2))$. Each $\mathcal{E}_k$ is evaluated as two matrix products, and the $r^2 \times r^2$ matrix is never formed. States are rescaled after every core and the logarithms of the scale factors are accumulated. Every pair marginal is exact in the same way:

$$p_{\mathcal{A}}^{(ij)}(x_i, x_j) = \frac{1}{Z}\,\big\langle \mathcal{E}_{>j}(\cdot),\; G_j(x_j)^{\mathsf{T}}\big[\mathcal{E}_{i+1:j-1}\big(G_i(x_i)^{\mathsf{T}}\mathcal{E}_{<i}(1)G_i(x_i)\big)\big]G_j(x_j)\big\rangle, \tag{7}$$

where $\mathcal{E}_{<i}$ is the left product of transfers and $\mathcal{E}_{>j}$ the right one. Every panel labelled "ft-de" in the figures is (7) evaluated on a grid, with no sampling or smoothing (`squared_pair_grid` in `ftde.marginals`). Pointwise evaluation of g costs $O(dr^2n)$.

Remark 2 (Why Chebyshev). The Gram matrices and moments (5) are exact and cheap. Chebyshev coefficients decay geometrically for analytic amplitudes, and chebdGE already builds cross approximations directly in this format. B-splines, as in TTDE, would give banded Gram matrices but piecewise-polynomial derivatives, and the smoothness functionals of §5 are more natural in a global polynomial basis. The price is the global support of T_i: a localized error in coefficients spreads over the whole axis. This is one mechanism for the ringing discussed in §8.3.

3 Fitting without noise

3.1 Objective and optimizer coordinates

With training data $\{X_i\}$ in Ω the objective is

$$\mathcal{L}(\mathcal{A}) = -\frac{1}{N}\sum_{i=1}^{N} \log g_{\mathcal{A}}(X_i)^2 + \log Z(\mathcal{A}), \tag{8}$$

minimized over the fixed-rank, fixed-degree coefficient set. The problem is nonconvex. Minibatch gradients come from JAX automatic differentiation of (8), whose ingredients are all polynomial evaluations and transfer contractions.

Physical (L^2) coordinates. Adam [12] rescales each coordinate separately, so its behaviour depends on the parametrization. Chebyshev coefficients are not L^2-orthonormal ($M_k \neq I$), and a unit step in a high-degree coefficient changes g much more in some directions than in others. We therefore optimize $B_k = A_k L_k$ (acting on the mode index), where $M_k = L_k L_k^{\mathsf{T}}$ is a Cholesky factor. In these coordinates the Euclidean inner product of mode vectors equals the $L^2([a_k, b_k])$ inner product of the univariate functions. The objective is unchanged. A *Sobolev* variant uses the metric $M_k + \lambda w_k D_k^{(s)}$, with $D_k^{(s)}$ the derivative Gram matrix of §5.1. This also changes only the optimizer coordinates, and it helps high-rank noisy fits whose high-order directions are stiff.

Schedule. Adam, with global-norm gradient clipping at 1, a cosine learning-rate decay to 10% of its initial value, and minibatches drawn with replacement. Held-out NLL (or held-out NLL plus penalty) is evaluated every 400 steps, and the best checkpoint is returned. Typical settings: 8,000 steps, learning rate 10^{-3}, batch 2048 (noise-free); $5 \cdot 10^{-4}$, batch 256 (noisy).

3.2 Initialization by chebdGE cross approximation

Let $\hat{f}$ be a Gaussian KDE of (a 5,000-point subsample of) the training data, with a diagonal bandwidth chosen by least-squares cross-validation (LSCV) [2, 17] as a scalar multiple of Scott's rule [18]. We run chebdGE's TT-qTensGE on $\sqrt{\hat{f}}$ (up to a scale) with the target rank and 25 modes. It builds each TT core by Gaussian elimination on the stage function $F_k(\alpha, x_k; x_{>k})$ with pivots found by a multi-start search, and represents left indices by prefix points. This is a continuous analogue of TT-cross [14] and of the Gaussian-elimination constructors of Chebfun2/3 [9, 20]. This gives an initial amplitude in the right functional class that is already a sensible density, unlike random or rank-one starts.

Remark 3. When the target is a thin tube in many dimensions (§8.3), the LSCV KDE is so concentrated that the pivot search can miss it entirely and return a rank-one, essentially empty TT. The implementation then broadens only the initializer bandwidth ($\times 2, \times 4$, then a quarter of the box). A candidate is accepted once all ranks exceed one and every observation has finite initial log density.

3.3 Capacity continuation

Rank and degree are increased by continuation. The modes are zero-padded, the TT is left-orthonormalized in physical coordinates (QR of each unfolded B_k, pushing R into the next core), and the cores are embedded in larger ranks with a small random perturbation of relative size 10^{-3} in the new blocks. The represented density changes only by that perturbation. The perturbation is necessary: if the new rank blocks are exactly zero, the gradient with respect to them vanishes identically, because every term involving a new direction contains another zero block, and the optimizer never uses them.

4 Deconvolution: the exact convolved likelihood

4.1 Model and ill-posedness

Under (1) the observation density of the model is

$$q_{\mathcal{A}}(y) = (U * p_{\mathcal{A}})(y) = \frac{1}{Z(\mathcal{A})} \int_{\Omega} U(y - x)\, g_{\mathcal{A}}(x)^2 \, \mathrm{d}x, \tag{9}$$

and we minimize $-\frac{1}{N}\sum_i \log q_{\mathcal{A}}(Y_i)$ (plus a penalty) over $\mathcal{A}$. Every noisy observation is retained, including those outside Ω. The latent box Ω is the 0.5–99.5% quantile box of the noisy data padded by 5%. Since $\hat{q} = \hat{U}\hat{p}$, directions of p at frequencies where $\hat{U}$ is small are barely visible in q. For Gaussian (supersmooth) noise the minimax rates for p are only logarithmic in N [3, 6, 19]. Two consequences drive the design: held-out *noisy* likelihood is nearly flat across candidates whose latent densities differ visibly, and regularization is indispensable.

4.2 Tensor-train noise kernels

We assume U is (represented by) a Chebyshev TT on a *difference box* $W = \prod_k [u_k, v_k]$: $U(z) = H_1(z_1) \cdots H_d(z_d)$, $H_k(z) = \sum_{l<m_k} C_k^{(l)} T_l(\sigma_k(z)) \in \mathbb{R}^{s_{k-1}\times s_k}$, with σ_k the affine map of $[u_k, v_k]$ to $[-1, 1]$. For observations in $\Omega^{\mathrm{obs}} = \prod [c_k, e_k]$ the window condition is

$$u_k \le c_k - b_k, \qquad v_k \ge e_k - a_k \qquad (k = 1, \ldots, d), \tag{10}$$

so that every difference $y_k - x_k$ that occurs lies in $[u_k, v_k]$. Examples used here:

- diagonal Gaussians: $s_k = 1$, each H_k a Chebyshev interpolant accurate to 10^{-11} relative;
- Gaussian mixtures $\sum_c w_c \mathcal{N}(0, \Sigma_c)$ with block-diagonal Σ_c: the block-diagonal TT sum of the components, with ranks adding. The cryo-EM kernel is a 4-component diagonal mixture obtained by clustering the supplied per-image variances;
- correlated 2×2 blocks: interpolate the bivariate Gaussian on a 513×513 tensor Chebyshev grid, take a 2D DCT, and truncate the SVD of the coefficient matrix (relative tolerance 10^{-13}). This yields two cores with an inner rank of about 28 for $\rho = 0.6$.

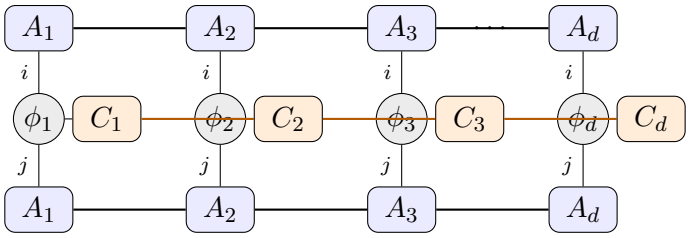


Figure 1: Tensor network for $Z\,q(y)$ in (11). Two copies of the amplitude TT (blue) and the noise TT (orange) meet at the per-axis moment tensors $\phi_k(y_k)$. These are exact polynomials in y_k, precomputed once. The contraction proceeds left to right with an $r \times r \times s$ state.

4.3 Exact evaluation

Proposition 4 (Exact convolved density)**.** *Let g be a Chebyshev TT on Ω with n_k modes and U a Chebyshev TT on W with m_k modes satisfying (10). For $y \in \Omega^{\mathrm{obs}}$ and $s \le 2n_k - 2$, define the* moment functions

$$\phi_k^{\beta\delta}(y_k; s) = \int_{a_k}^{b_k} H_k^{\beta\delta}(y_k - x)\, T_s(\tau_k(x))\, \mathrm{d}x.$$

Then

$$\begin{aligned} Z\,q(y) &= \Phi_d(y_d) \circ \cdots \circ \Phi_1(y_1)(1), \\ \Phi_k(y_k)(S)[\gamma, \gamma', \delta] &= \sum_{\alpha,\alpha',\beta} \sum_{i,j} S[\alpha, \alpha', \beta]\, A_k^{(i)}[\alpha, \gamma] A_k^{(j)}[\alpha', \gamma']\ \tfrac{1}{2}(\phi_k^{\beta\delta}(y_k; i{+}j) + \phi_k^{\beta\delta}(y_k; |i{-}j|)), \end{aligned} \tag{11}$$

where the states S are $r_{k-1} \times r_{k-1} \times s_{k-1}$ arrays. Moreover:

- *(a) for fixed y_k, $\phi_k(y_k; s)$ is computed exactly by Gauss–Legendre quadrature on $[a_k, b_k]$ with $Q_k = n_k + \lfloor (m_k - 1)/2 \rfloor$ nodes;*
- *(b) $y_k \mapsto \phi_k^{\beta\delta}(y_k; s)$ is a polynomial of degree at most $m_k - 1$, so it equals its m_k-point Chebyshev interpolant on $[c_k, e_k]$.*

Proof. Formula (11) is Fubini applied to $U(y-x)g(x)^2 = \prod_k H_k(y_k - x_k) \otimes G_k(x_k) \otimes G_k(x_k)$, followed by the linearization $T_i T_j = \frac{1}{2}(T_{i+j} + T_{|i-j|})$. For (a), the window condition puts $y_k - x \in [u_k, v_k]$ for all $x \in [a_k, b_k]$, so the integrand is a polynomial in x of degree $\le (m_k - 1) + (2n_k - 2) \le 2Q_k - 1$ (check both parities of m_k). For (b), $H_k^{\beta\delta}(y - x) = \sum_{l < m_k} c_l\, T_l(\sigma_k(y - x))$ is a polynomial of degree $\le m_k - 1$ in y for every x, and integration in x preserves this. □

The implementation (`TTConvolution.prepare`) evaluates (a) at the m_k Chebyshev points of $[c_k, e_k]$ once, before optimization, and stores the resulting *moment cores*, the Chebyshev coefficients of $\phi_k(\cdot; s)$ for $s \le 2n_k - 2$. During training, each observation needs only a Chebyshev evaluation of these cores and the contraction (11). The latter is three products per core, costing $O(r^3 s\, n + r^2 s^2 n^2)$ per observation and core. Block-diagonal (mixture) kernels are contracted block by block. The normalizer Z is shared with the noise-free model, and $\int q = \int_W U$. Every fit ends with an independent check at eight validation points against a direct (unprepared) evaluation at twice the quadrature order, and the fit is rejected if any log density differs by more than 10^{-7}. The largest difference observed across all gallery fits is $2.1 \cdot 10^{-12}$.

Remark 5 (What is and is not exact). The only approximation in the likelihood is the representation of the prescribed noise by the Chebyshev TT U on the window W. That representation is checked against the analytic kernel (to 10^{-9} of its peak for the kernels used here) and can be made as accurate as desired by adding modes. In particular there is no sampling of the noise and no approximation of g^2 by a lower-rank TT. Chebyshev approximations of Gaussians can have tiny negative tails. Plots clip these, and the clipped mass is recorded and required to be below 10^{-5}.

5 Regularization

5.1 Amplitude Sobolev seminorms

For $s \in \{1, 2\}$ and length scales $\ell_k > 0$ define

$$R_s(\mathcal{A}) = \sum_{k=1}^{d} \ell_k^{2s} \, \frac{\|\partial_k^s g_{\mathcal{A}}\|^2_{L^2(\Omega)}}{\|g_{\mathcal{A}}\|^2_{L^2(\Omega)}}. \tag{12}$$

It is invariant under gauge, scale and sign, since it depends only on $p_{\mathcal{A}}$. For $s = 1$,

$$\frac{\|\partial_k g\|^2}{\|g\|^2} = \int (\partial_k \sqrt{p})^2 = \frac{1}{4} \int \frac{(\partial_k p)^2}{p}, \tag{13}$$

one quarter of the location Fisher information of p along axis k. So R_1 penalizes exactly the quantity that a noisy likelihood cannot see well. The numerator is computed exactly by replacing M_k with the derivative Gram matrix $D_k^{(s)} = \Delta_k^s M_k (\Delta_k^s)^{\mathsf{T}}$ in one factor of the transfer recursion, where Δ_k is the Chebyshev differentiation matrix scaled by $2/(b_k - a_k)$. With (N_k, E_k) the running (numerator, norm) states, $N_k = \mathcal{E}_k(N_{k-1}) + \ell_k^{2s} \mathcal{D}_k(E_{k-1})$ and $E_k = \mathcal{E}_k(E_{k-1})$. In deconvolution we take ℓ_k to be the standard deviation of the k-th univariate marginal of U, computed from the noise TT itself, and use $s = 2$ ("amplitude H^2"). Mixed derivatives are omitted.

5.2 Marginal Fisher-visibility filter

The Sobolev penalty damps every high-frequency direction equally. The *visibility filter* instead damps the directions the noise actually hides. Fix an axis k, a reference marginal density ρ on $[a_k, b_k]$ (the *pilot*, §5.3), and the univariate noise marginal u of U. Consider amplitude perturbations $\sqrt{\rho} \mapsto \sqrt{\rho} + th$ with $h \perp \sqrt{\rho}$ in L^2 and $\|h\| = 1$. The complete-data Fisher information of t, divided by four, equals $\|h\|^2 = 1$. The *observed* (noisy) one is

$$\mathcal{F}(h, h) = \int \frac{\big(u * (\sqrt{\rho}\, h)\big)^2}{u * \rho} \, \mathrm{d}y \in [0, 1], \tag{14}$$

after projecting out the normalization direction. It lies in $[0, 1]$ because the noisy score is a conditional expectation of the clean score. Discretize h in the L^2-orthonormal Chebyshev basis restricted to $\sqrt{\rho}^{\perp}$, and diagonalize $\mathcal{F} = \sum_m \lambda_m v_m v_m^{\mathsf{T}}$. With a visibility floor $\alpha > 0$ the per-axis penalty operator is

$$P_k = \sum_m \frac{\alpha}{\alpha + \lambda_m} \, v_m v_m^{\mathsf{T}} \qquad (\text{and } P_k \sqrt{\rho} = 0), \tag{15}$$

mapped back to Chebyshev coefficients. The functional $R_F(\mathcal{A}) = \sum_k \langle g, (P_k)_{\text{axis } k}\, g \rangle / \|g\|^2$ is computed by the same two-state recursion as R_s, with P_k in place of $D_k^{(s)}$. Nearly invisible directions ($\lambda \ll \alpha$) receive weight ≈ 1, well-observed ones ($\lambda \gg \alpha$) weight $\approx \alpha/\lambda$, and the reference itself is free. The full correlated U still enters the likelihood. Only the regularizer uses marginal geometry, and it needs no Gaussian assumption on U.

5.3 Pilot marginals

Pilots are exponential-family marginals $\log \rho_k = \sum_{j=1}^{m} \theta_{kj} T_j \circ \tau_k$ with $m \in \{2, 4, 6, 8\}$, either shared across axes (in standardized coordinates) or separate. They are fitted by the binned noisy marginal likelihood through the univariate noise marginal (L-BFGS, 1,600 bins, at most 1,200 iterations), from a moment-corrected Gaussian start. Degree and sharing are chosen by the rule of §6, with the parameter count as the complexity.

6 Model selection from noisy observations

Let $\ell_c(y_v)$ be the held-out log density of candidate c at validation point y_v ($v = 1, \ldots, V$), and $c^\star = \arg\min_c(-\frac{1}{V}\sum_v \ell_c(y_v))$. For each c form the paired excesses $e_{cv} = \ell_{c^\star}(y_v) - \ell_c(y_v)$, their mean $\bar{e}_c$, and a cluster-robust standard error SE_c (clusters are MCMC chains for Ginzburg–Landau, single observations otherwise). The eligible set is $\{c : \bar{e}_c \le \mathrm{SE}_c\}$, and among eligible candidates we pick the one with the smallest value of a *common* roughness functional (the Fisher penalty with a common filter, or R_2 with common length scales). This is the one-standard-error rule [10] applied to paired differences. It is a heuristic, not a post-selection confidence statement, and the same validation split is reused across stages. As the experiments show (§8.3), it is also the weakest link. Candidates that differ visibly in the latent density can be indistinguishable in noisy NLL.

7 The two recipes

Algorithm 1 Gallery recipe for Ginzburg–Landau (`scripts/refresh_density_fits.py`)

1: Latent box: training min/max +10% (noise-free) or noisy 0.5–99.5% quantiles +5% (noisy).
2: LSCV bandwidth; TT-qTensGE of $\sqrt{\mathrm{KDE}}$ at rank r (8 for $d \le 8$, 10 for noisy or $d \le 14$, else 14), 21 (noise-free) or 25 (noisy) modes, 64 starts.
3: **if** noise-free **then**
4: Adam on (8), 8,000 steps, lr 10^{-3}, batch 2048; checkpoint by validation NLL.
5: **else**
6: Warm-up: Adam on noisy NLL + $3 \cdot 10^{-5} R_2$, 8,000 steps, lr $5 \cdot 10^{-4}$, batch 256.
7: Pilots ρ_k (degrees 2–8, shared/separate) selected by the paired one-SE rule.
8: Pad to 33 modes (more if the pilot amplitude is under-resolved); build R_F with $\alpha = 0.01$.
9: For $\lambda \in \{0.1, 0.3, 1\}$: continue from the warm-up on noisy NLL $+\lambda R_F$, 8,000 steps.
10: Select among warm-up and the three Fisher fits: paired one-SE, then minimum common R_F.
11: **end if**

Algorithm 2 High-capacity recipe for cryo-EM and swirl–pretzel (`refine_cryoem_continuity.py`, `swirl_fits.py`)

1: Base fit as in Algorithm 1 (init rank 16 for swirl), without the Fisher stage.
2: For $r \in \{32, 64\}$: continue at 49 modes (§3.3). Noisy: $\lambda \in \{10^{-4}, 3 \cdot 10^{-4}\}$ times R_2, Sobolev preconditioning, 3,200 steps, lr $2 \cdot 10^{-4}$, batch 256, checkpoints by validation NLL $+\lambda R_2$.
3: Noise-free: minimum validation NLL. Noisy: paired one-SE, then minimum common R_2.

For rank 64 the backward pass of (11) is memory-bound: one intermediate has $B \cdot r^2 s\, n$ entries. The same 256-row minibatch gradient is therefore accumulated over four chunks, which gives an

identical gradient.

8 Experiments

All experiments use one GPU (NVIDIA A100 or H100). Truth, clean samples and state labels enter only after fitting and selection. The error rel L^2 is always full-dimensional. For Ginzburg–Landau, $\int \hat{p}^2$ is Gauss–Legendre quadrature of the degree-$4(n-1)$ amplitude polynomial, and $\int p\hat{p}$ and $\int p^2$ come from the transfer-operator recursion of the chain. For the swirl–pretzel density they are path quadratures (§8.3). Every quadrature is refined until it changes by less than 10^{-7} relative.

8.1 Ginzburg–Landau

The target is the discretized Ginzburg–Landau field of [16, Simulation IV],

$$p(x) \propto \exp\Big(-\beta\Big[\sum_{i=0}^{d} \tfrac{\lambda}{2}\big(\tfrac{x_{i+1}-x_i}{h}\big)^2 + \sum_{i=1}^{d} \tfrac{1}{4\lambda}(1-x_i^2)^2\Big]\Big), \quad x_0 = x_{d+1} = 0,\ \beta = \tfrac{1}{8},\ \lambda = 0.02,\ h = \tfrac{1}{d+1}.$$

It has double-well marginals coupled along a chain, so every pair marginal is a set of four blobs. Samples come from 200 single-site Metropolis chains; there are 100,000 training and 2,048 validation samples. Two noise families are scaled by 1.5: a *paired* Gaussian with standard deviation 0.83 and correlation 0.5 within coordinate pairs, and an isotropic *mixture* $0.7\,\mathcal{N}(0, 0.45^2 I) + 0.3\,\mathcal{N}(0, 1.35^2 I)$. The noise is comparable to the distance between the wells at ± 1. The recipe is Algorithm 1.

d	noise-free	paired Gaussian noise	Gaussian-mixture noise
6	0.081	0.095	0.131
10	0.109	0.141	0.136
14	0.132	0.231	0.169
20	0.161	0.244	0.236

Table 1: Ginzburg–Landau, $N = 100{,}000$: rel L^2 of the recovered latent density (Algorithm 1).

The errors in Table 1 grow slowly with d. Figure 2 shows the qualitative behaviour: the deconvolved estimate recovers four well-separated modes from data in which they are not visible. The GL density is close to low rank and its pair marginals are nearly separable blobs. These are favourable conditions for a squared Chebyshev TT, which is why we also test the harder geometry of §8.3.

Fisher-visibility spectra. Figure 4 shows the eigenvalues λ_m of (14) for the $d = 10$ mixture case. They decay roughly geometrically, $\lambda \approx 0.63, 0.18, 0.07, 0.02, 0.007, \ldots$, so on each axis only four or five amplitude directions are observed above the floor $\alpha = 0.01$. The filter leaves those almost free and damps the rest.

8.2 Sample-size convergence

To measure the finite-N behaviour we reran the recipes *unchanged* for $N \in \{10^3, 3\cdot 10^3, 10^4, 3\cdot 10^4, 10^5, 3\cdot 10^5\}$: GL in $d = 10$, noise-free and with mixture noise (Algorithm 1, training seeds 0 and 1), and the swirl–pretzel in $d = 4$, noise-free and noisy (Algorithm 2 with rank-32 continuations only, one seed). Rank, degree, step counts and penalty grids do not depend on N, and the validation

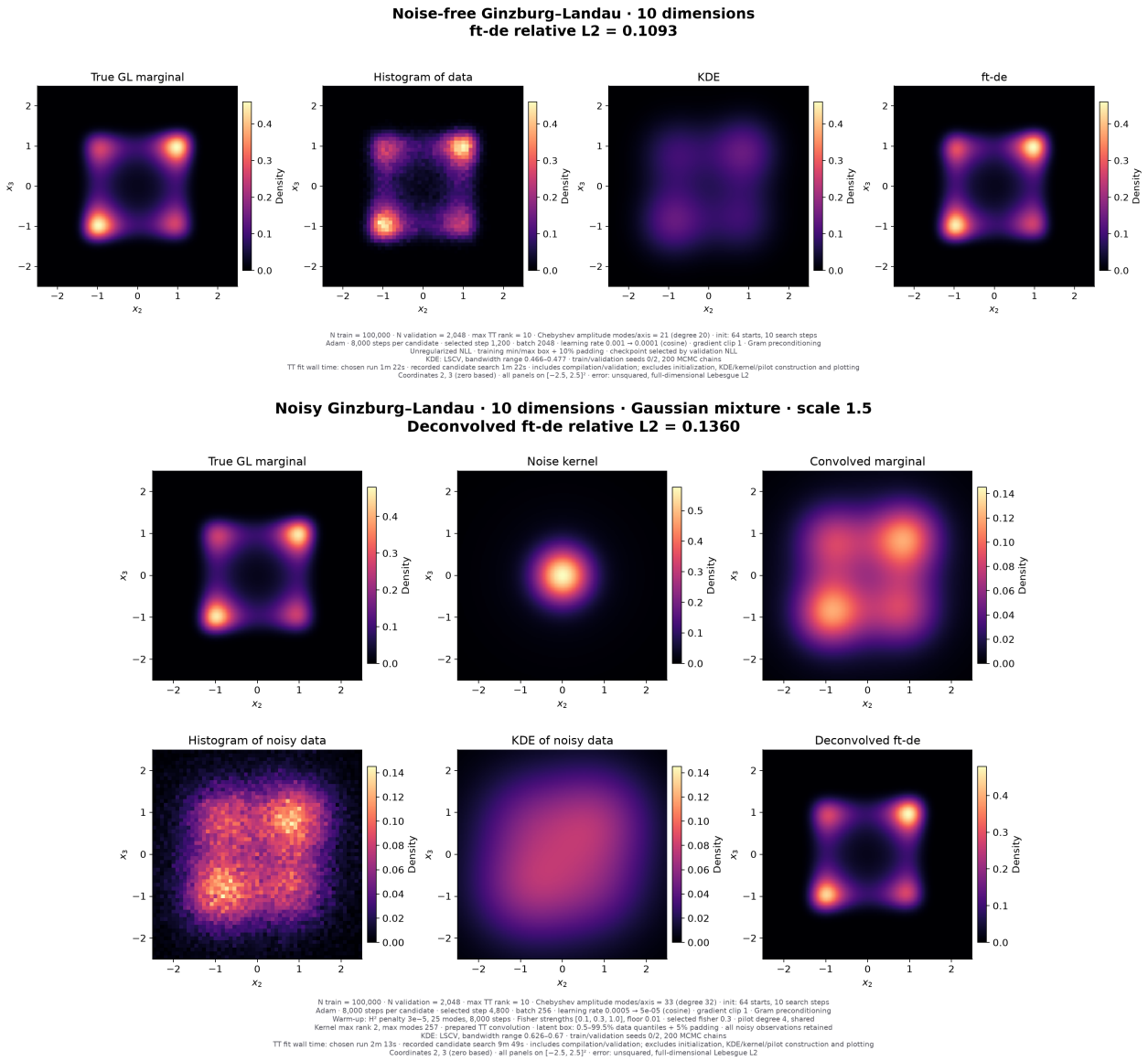


Figure 2: Ginzburg–Landau, $d = 10$, coordinates $(2, 3)$. Top: noise-free (truth, training histogram, KDE, ft-de). Bottom: mixture noise. Top row: truth, noise kernel, true convolved marginal. Bottom row: histogram and KDE of the noisy data, and the deconvolved latent ft-de, with the same color scale as the truth. The recovered density separates the four wells, including their unequal weights, although the noisy data show only two diffuse lobes.

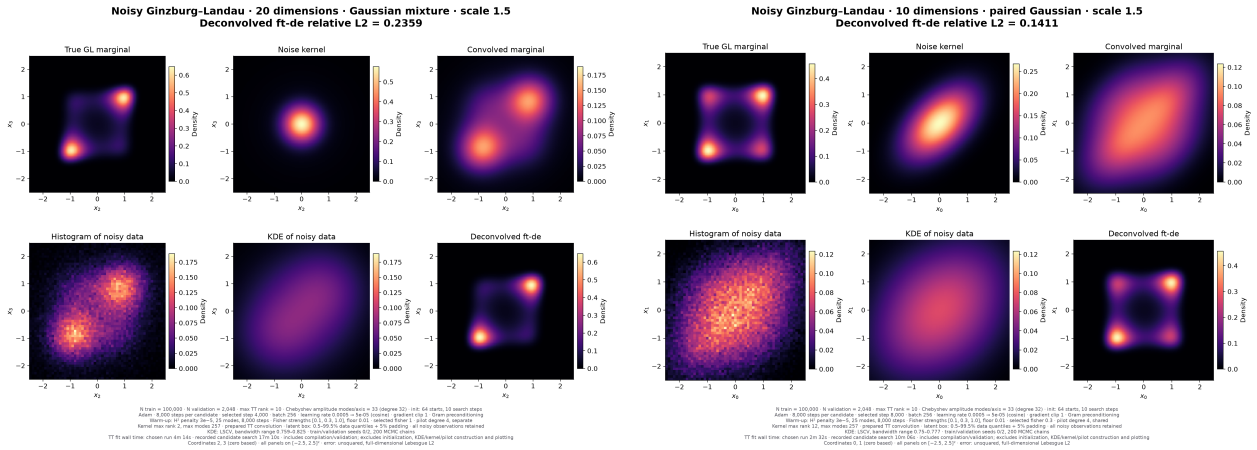


Figure 3: Left: GL, $d = 20$, mixture noise (rel $L^2 = 0.236$). Right: $d = 10$, correlated paired noise (0.141).

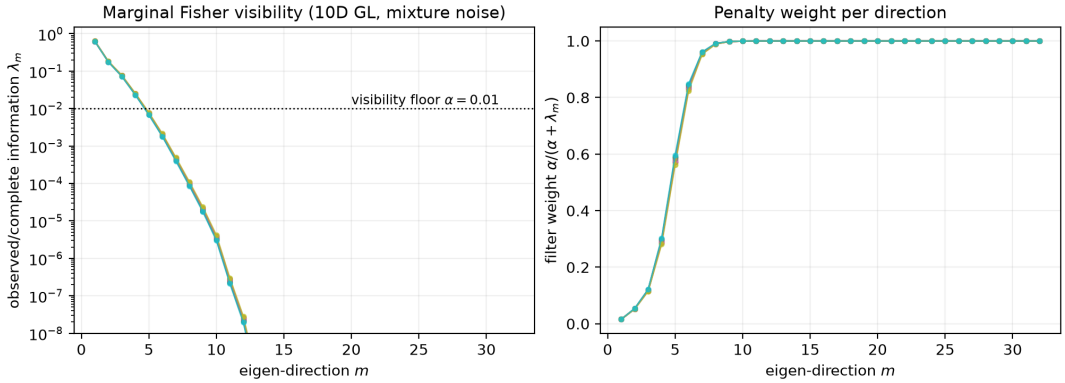


Figure 4: Marginal Fisher-visibility eigenvalues (14) (one curve per axis) and the resulting penalty weights (15), GL $d = 10$ with mixture noise.

sets are fixed (2,048 GL and 4,096 swirl observations), so this measures the recipe as used, not an estimator tuned to each N. The baseline is a Gaussian KDE whose LSCV bandwidth is rescaled with Scott's exponent $N^{-1/(d+4)}$. In the noisy cases it is the KDE of the *noisy* observations. It estimates q, not p, so it shows what is lost without deconvolution. All errors are full-dimensional rel L^2 against the clean latent density (for the GL KDE, $\int p\,\widehat{p}_{\mathrm{KDE}}$ is a Monte Carlo estimate over 200,000 exact samples, with standard error below 10^{-3}).

1. *Every ft-de curve decreases monotonically in N* (Figure 5, Table 2). The log–log slopes between $N = 10^4$ and $3 \cdot 10^5$ are -0.54 (GL, noise-free), -0.20 (GL, noisy), -0.33 (swirl, noise-free) and -0.23 (swirl, noisy). Over the last interval, 10^5 to $3 \cdot 10^5$, they are -0.40, -0.05, -0.12 and -0.26: the noisy GL and noise-free swirl curves level off, the other two do not.

2. *Noise-free GL* follows the $N^{-1/2}$ guide from 10^4 to $3 \cdot 10^5$ (0.44 to 0.071). The KDE barely moves (0.97 to 0.85, slope -0.03), well below even the asymptotic kernel rate $N^{-2/(d+4)} = N^{-1/7}$: in ten dimensions it is still pre-asymptotic. For $N \le 3 \cdot 10^3$ the unregularized noise-free recipe is worse than the zero function (rel $L^2 = 1.3$–2.3): with about $1.7 \cdot 10^4$ coefficients, checkpointing by validation NLL does not prevent overfitting. The overfitted fits are too peaked rather than misplaced: $\int \hat{p}^2$ is 2.7–6 times $\int p^2$, while $\int p\hat{p} \approx \int p^2$. A penalty or a smaller rank at small N

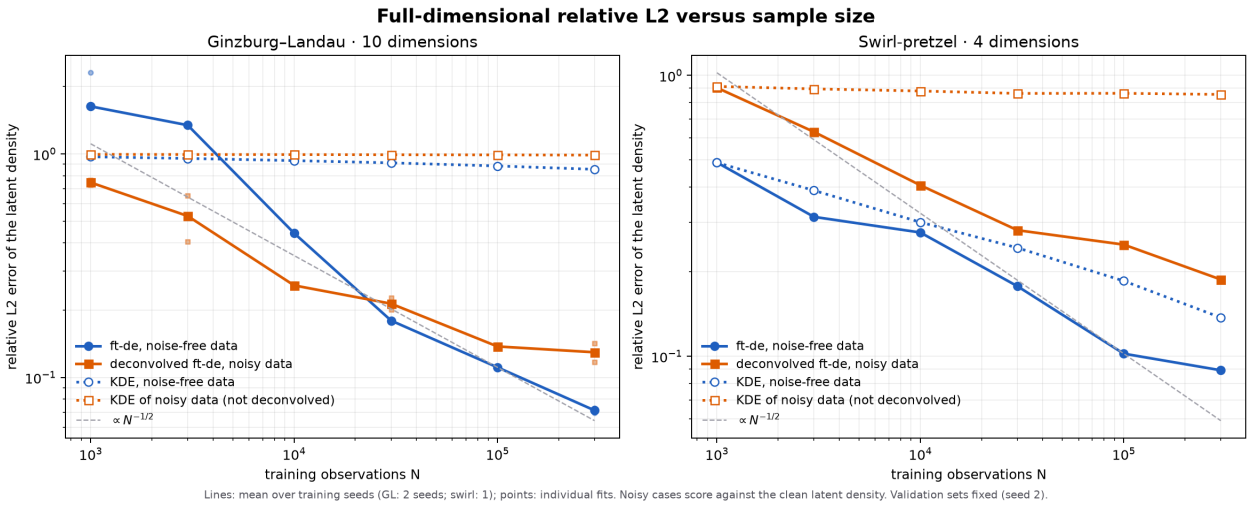


Figure 5: Full-dimensional rel L^2 of the latent density against the number of training observations (log–log). Solid: ft-de; dotted: Gaussian KDE (in the noisy cases, of the noisy data). Lines are means over training seeds, points individual fits. The dashed guide has slope $-\frac{1}{2}$, anchored at the noise-free fit at $N = 10^5$.

N	10^3	$3 \cdot 10^3$	10^4	$3 \cdot 10^4$	10^5	$3 \cdot 10^5$
GL, noise-free (ft-de)	1.633	1.345	0.442	0.180	0.111	0.071
GL, noise-free (KDE)	0.971	0.955	0.934	0.912	0.884	0.854
GL, mixture noise (ft-de)	0.746	0.528	0.258	0.214	0.138	0.130
GL, mixture noise (KDE)	0.996	0.995	0.994	0.992	0.990	0.988
swirl, noise-free (ft-de)	0.490	0.314	0.276	0.178	0.102	0.089
swirl, noise-free (KDE)	0.489	0.390	0.300	0.243	0.186	0.137
swirl, noisy (ft-de)	0.904	0.630	0.407	0.281	0.250	0.188
swirl, noisy (KDE)	0.914	0.896	0.880	0.863	0.864	0.856

Table 2: Data of Figure 5: rel L^2, mean over training seeds (two for GL, one for the swirl).

would be needed.

3. *Noisy GL.* The deconvolved estimate beats the KDE of *clean* data at every N, and at $N \le 3 \cdot 10^3$ it also beats the noise-free ft-de fit, presumably because the noisy recipe is regularized. Its error falls from 0.75 to 0.13 but levels off above 10^5 (0.138 and 0.130; seeds 0.142 and 0.117 at $3 \cdot 10^5$). We cannot say from this sweep whether the floor is statistical or computational. The step budget is fixed (8,000 steps of 256 observations, about seven passes over the data at $N = 3 \cdot 10^5$), and so are the rank (10) and the penalty grid.

4. *Noise-free swirl.* ft-de and KDE tie at $N = 10^3$ (0.49). At every larger N ft-de is more accurate, by a factor of 1.1–1.8 (largest at 10^5), and it reaches 0.089 at $3 \cdot 10^5$. The flattening between 10^5 and $3 \cdot 10^5$ (0.102 to 0.089) is consistent with a rank-32 capacity floor (the remark in §8.3 predicts that ranks of several tens are needed), but the sweep did not try other ranks.

5. *Noisy swirl.* Deconvolution gains nothing at $N = 10^3$ (0.90 against the noisy KDE's 0.91). After that, the error falls to 0.19 at $3 \cdot 10^5$, while the noisy KDE stays at 0.86–0.90. Noise costs roughly a factor of ten in sample size here: the deconvolved fit at $3 \cdot 10^5$ matches the noise-free

fit at about $3 \cdot 10^4$. The sweep's value at $N = 10^5$ (0.250) is above the 0.216 of Table 3 because the sweep does not try rank 64.

These slopes are empirical and cover only two and a half decades. They are not in conflict with the logarithmic minimax rate for supersmooth noise [6]. That rate is a worst case over a smoothness class, whereas both targets are close to low rank and analytic, and at fixed capacity the relevant question is how fast the variance of a finite-dimensional fit decays.

8.3 A continuous benchmark: the swirl–pretzel tube

The deconvolved cryo-EM marginals (§8.4) look *beaded*: bright blobs strung along curved arms. To test whether the estimator fragments continuous ridges, we built a target whose truth is a continuous, curved ridge with the same noise-to-width ratio as cryo-EM. A latent path parameter $t \in [0, 1]$ with density $w(t) \propto 1 + 0.6 \sin(4\pi t + 0.4)$ is embedded as

$$X = \gamma(t) + 0.14\,\xi, \qquad \xi \sim \mathcal{N}(0, I_d),$$

where (γ_0, γ_1) is an Archimedean spiral (1.6 turns; the *swirl*), (γ_2, γ_3) a trefoil with three self-crossings (the *pretzel*), and γ_k for $k \geq 4$ are slow sinusoids plus drift. This is the continuous analogue of the cryo-EM latent space, one path in $\mathbb{R}^d$, but without discrete states. Its density $p(x) = \int_0^1 w(t) \prod_k \mathcal{N}(x_k; \gamma_k(t), 0.14^2)\,\mathrm{d}t$, pair marginals, $\int p^2$ (a double path integral), and the exact noisy density (the path integral of Gaussians with covariance $\Sigma_c + 0.14^2 I$) are all computed by path quadrature with error $\approx 10^{-10}$. For $\int p\hat{p}$ we use $\int w(t) \int \hat{g}^2(x) \prod_k \mathcal{N}(x_k; \gamma_k(t), 0.14^2)\,\mathrm{d}x\,\mathrm{d}t$. For each path node the inner integral is a product of Gaussian-weighted polynomials, exact under Gauss–Hermite quadrature, so it is a transfer contraction as in (6). We cross-checked it against 200,000-sample Monte Carlo.

Remark 6 (Rank of a tube, and why beads are the signature of too little rank). For every split $x = (x_{\leq k}, x_{>k})$, $p(x) = \int_0^1 w(t)\, a_k(x_{\leq k}; t)\, b_k(x_{>k}; t)\,\mathrm{d}t$ with a_k, b_k products of Gaussians centred on the path. Every unfolding of p is therefore a continuous superposition of rank-one terms indexed by t. A rank-r TT is akin to an r-node quadrature of this integral. It is accurate once the nodes are closer than the tube width along the path, i.e. $r \gtrsim L_k/(0.14)$, where L_k is the length of the path projected onto the smaller of the two coordinate blocks. For the spiral alone $L \approx 14$, so ranks of several tens are needed. Squaring changes the constants but not the picture. With too little effective rank, the fit resembles a quadrature with too few nodes: about r blobs strung along the ridge, i.e. beads. Separable pieces aligned with the axes produce the straight segments seen below. This is a heuristic explanation, not a theorem, but it predicts what we observe: beads appear where information is weakest (faint path segments, noisy data) and in higher d, where the path is longer in every unfolding.

The noise is a sum of three Gaussians: $0.5\,\mathcal{N}(0, 0.22^2 I) + 0.3\,\mathcal{N}(0, \mathrm{diag}(0.45^2, 0.25^2, \dots)) + 0.2\,\mathcal{N}(0, \Sigma_\rho)$, where Σ_ρ has 2×2 blocks $0.5^2 \left[\begin{smallmatrix} 1 & \pm 0.6 \\ \pm 0.6 & 1 \end{smallmatrix}\right]$. Its kernel TT has ranks 30 within pairs and 3 between pairs. The per-axis noise standard deviation (0.30–0.37) is about 2.5× the tube width, the same ratio as the cryo-EM oracle below. The spiral's turns are 2.7–3.3 noise standard deviations apart. There are $N = 100{,}000$ i.i.d. training and 4,096 validation samples, and the fits use Algorithm 2.

Findings.

1. *No noise, no beads.* In all dimensions the noise-free fits are continuous (Figure 6), with ripple at most 3.7% for every searched candidate. Representation and optimization alone do not fragment

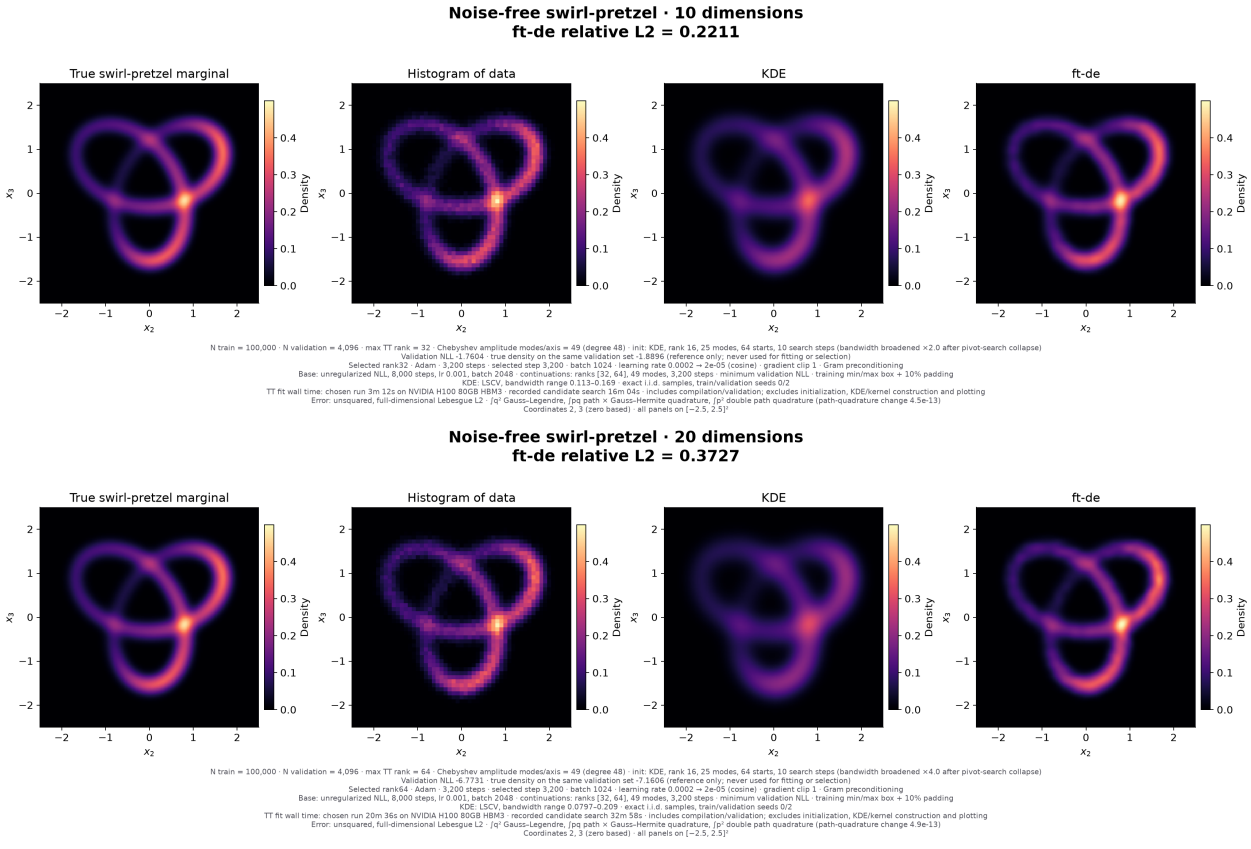


Figure 6: Noise-free swirl–pretzel, pretzel coordinates, $d = 10$ (top) and $d = 20$ (bottom). The recovered ridge is continuous, with no beads, and sharper than the KDE.

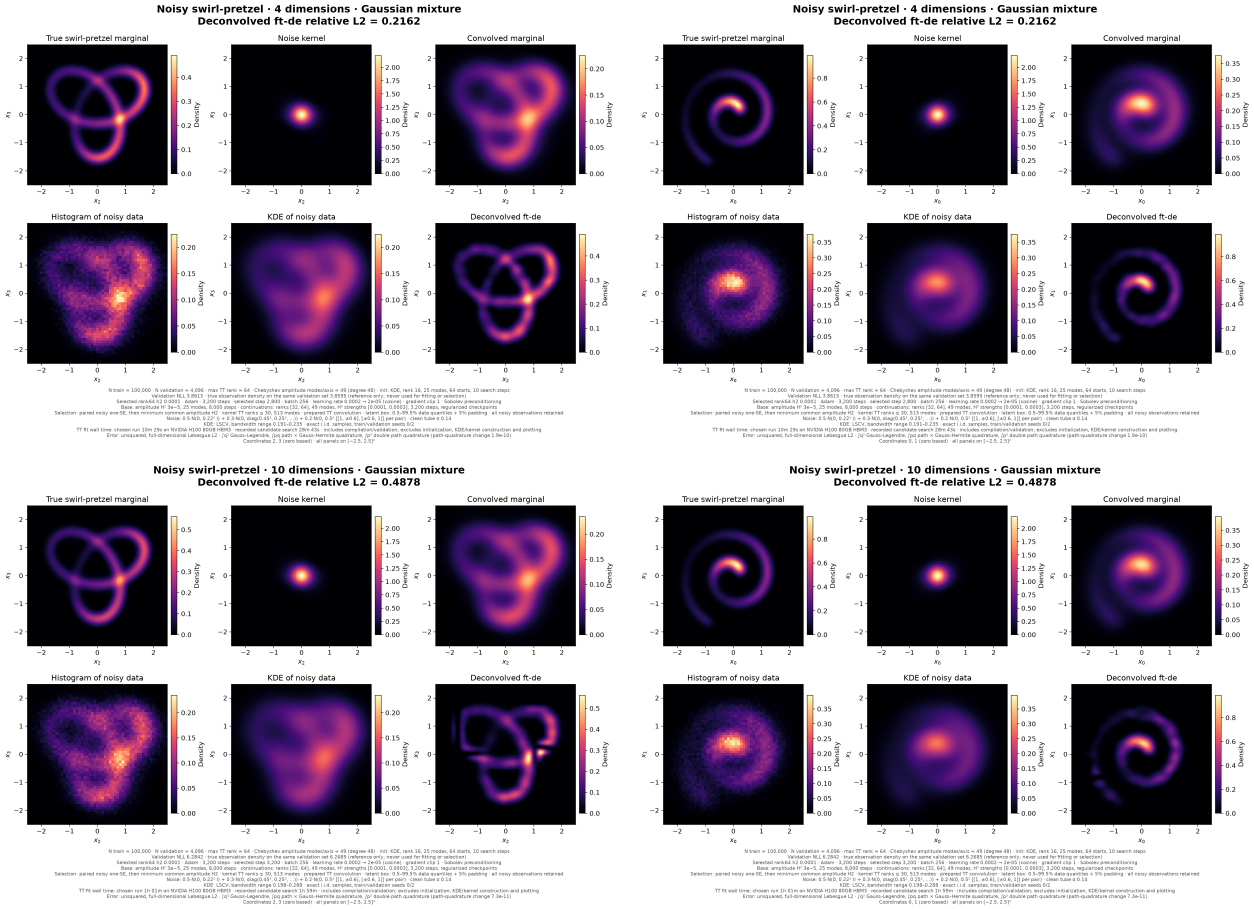


Figure 7: Noisy swirl–pretzel. Top: $d = 4$; the deconvolved density (bottom right of each panel) recovers the trefoil and the spiral, including the inner lobes that are invisible in the noisy histogram and KDE, with only faint ripples. Bottom: $d = 10$. The spiral's faint outer arm breaks into beads, and the pretzel has straight axis-aligned segments, a spurious detached blob near $(1.3, 0.1)$ and a missing crossing.

	d	selected	rel L^2	NLL gap to truth (nats)	ripple (0,1) / (2,3)
noise-free	4	rank 32	0.102	0.033 ± 0.004	0.015 / 0.032
	10	rank 32	0.221	0.129 ± 0.007	0.018 / 0.024
	20	rank 64	0.373	0.388 ± 0.013	0.015 / 0.028
noisy	4	rank 64, $\lambda = 10^{-4}$	0.216	0.0021 ± 0.0010	0.023 / 0.062
	10	rank 64, $\lambda = 10^{-4}$	0.488	0.0157 ± 0.0027	0.046 / 0.183
	20	rank 32, $\lambda = 3 \cdot 10^{-4}$ †	0.866	0.160 ± 0.008	0.032 / 0.088

Table 3: Swirl–pretzel results. The *NLL gap* is the paired mean, on held-out observations (noisy ones in the noisy rows), of $\log q_{\text{true}} - \log q_{\text{fit}}$. It is a reference that fitting can approach but not beat in expectation. *Ripple* isolates bead-like errors. The estimate minus the truth is evaluated along the projected true path, high-pass filtered (Gaussian filter in t, width 0.012, about 1.5 tube widths), and its norm is divided by that of the true profile. Sharp features of the truth itself (crossings, fold caustics) cancel in the difference. †In $d = 20$ the rank-64 noisy continuations were omitted for compute (an estimated ≈ 6 h each on a shared GPU), so selection is among the base fit and two rank-32 fits.

ridges. Rank-64, 49-mode fits show faint ripple along the ridge from the global Chebyshev basis, far below bead contrast.

2. *Deconvolution in $d = 4$ is nearly as good as the data allow.* The selected fit is within 0.0021 ± 0.0010 nats of the true noisy density, and its marginals are continuous (Figure 7, top).

3. *Deconvolution in $d = 10$ fragments ridges* (Figure 7, bottom; ripple on the pretzel 16–19%). The artifacts are already present in the rank-16 base fit, and every continuation inherits them (Figure 9). The noisy likelihood *does* detect that this fit is wrong: the truth scores 0.0157 ± 0.0027 nats better, about six standard errors. The failure is therefore not an identifiability limit at this N. The optimizer is stuck in a worse basin. Figure 10 repeats the $d = 10$ fit (rank 32, $\lambda = 3 \cdot 10^{-4}$) from three chebdGE initializations (different KDE subsamples and pivot searches). All three reach the same noisy NLL within noise (gaps to the truth 0.030, 0.033, 0.031, each ± 0.004) and the same latent error (rel $L^2 = 0.588, 0.588, 0.600$). The artifacts, however, sit in *different places*: a detached blob, vertical and horizontal streaks, or a broken spiral arm. The normalized-density average of the three has the same noisy NLL as its best member ($\Delta = 0.0003 \pm 0.0007$), yet its pretzel marginal is markedly closer to the truth (2D relative L^2 0.18 against 0.21–0.26). Artifact *locations* are therefore basin-dependent and invisible to the noisy likelihood. Agreement across independent initializations is a diagnostic the likelihood cannot provide.

4. *In $d = 20$* the pretzel marginal is again mostly continuous, with a broken upper-left lobe (Figure 8), but the fit is far from the truth in full dimension (rel $L^2 = 0.87$; NLL gap 0.160 ± 0.008 nats). With a fixed budget of steps and rank, the gap to the truth grows with d both with and without noise (Table 3).

5. *Held-out noisy NLL tracks latent error* (Figure 11). Across every searched candidate, in every dimension, a smaller gap to the true observation density goes with a smaller latent rel L^2. The only inversion, between the two rank-32 fits in $d = 10$, is within one standard error. The one-standard-error rule therefore selects sensibly among the candidates it is offered. Its weakness is that it only chooses among fits that share the base fit's topology.

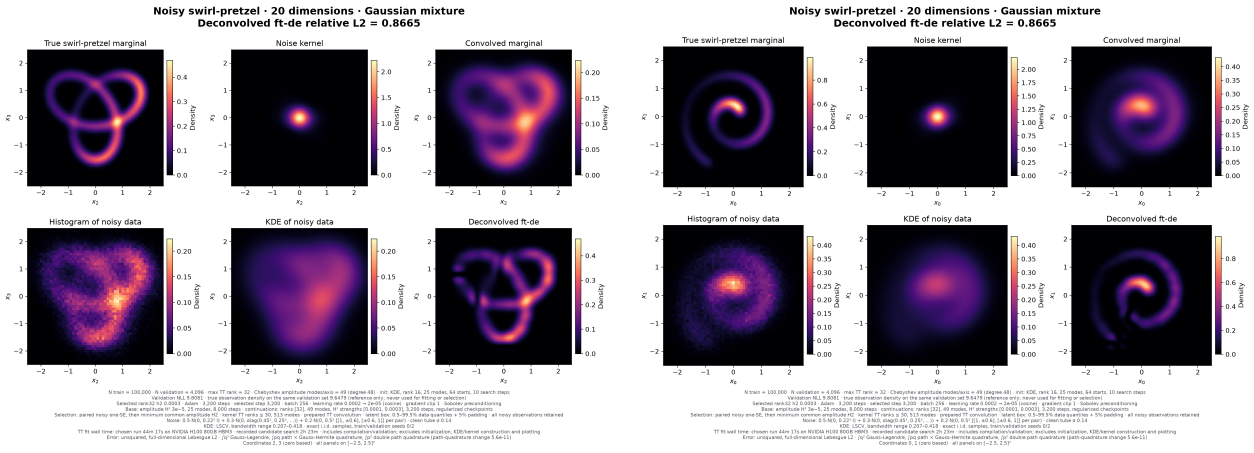


Figure 8: Noisy swirl–pretzel, $d = 20$ (rank 32, $\lambda = 3 \cdot 10^{-4}$).

6. *Smoothing trades beads for bias* (Figure 12). Raising the amplitude-H^2 strength in $d = 10$ from 10^{-4} to 10^{-2} removes the spiral beads (ripple $0.050 \to 0.008$) and halves the pretzel ripple ($0.19 \to 0.10$). But rel L^2 rises from 0.55 to 0.90 and the NLL gap from 0.030 to 0.109 nats. The pretzel's misplaced segments remain, only blurred. Beads are a visible but minor part of the error.

8.4 Cryo-EM latent space

The data are RECOVAR [7] latent embeddings of a simulated heterogeneous dataset, in 4, 10 and 20 dimensions (separately fitted embeddings), together with per-image posterior covariances. After dropping the 1% of images with the largest covariance trace, 99,000 remain: 94,904 for training and 4,096 for validation. Coordinates are standardized, and plots map back with the Jacobian. The prescribed noise is a 4-component mixture of diagonal Gaussians obtained by k-means clustering of the log per-image variances. It uses only the supplied metadata, never positions or labels. The current fits come from Algorithm 2 continued from earlier gallery fits: ranks 32 (4D), 64 (10D) and 48 (20D), 49 modes, H^2 strength 10^{-4} (4D) or $3 \cdot 10^{-4}$ (10D and 20D), Sobolev preconditioning. Held-out noisy NLLs are 3.531, 9.857 and 24.657.

Is the clean density beaded? The deconvolved marginals in Figure 13 are beaded. The simulation assigns every image to one of *50 discrete conformational states* (between 20 and about 4,900 images each), and these labels can be used, after all fitting, to build an oracle. For each state we take the mean of its noisy embeddings and the PSD part of (within-state covariance − mean supplied noise covariance). The oracle clean density is the resulting 50-component Gaussian mixture. We compare it with a *continuous-path counterfactual*: consecutive state means joined by segments, with interpolated weights and covariances. The oracle convolved with the noise reproduces the observation-space TT fit to 2D relative L^2 between 0.02 and 0.14, which validates both the oracle and the supplied noise model.

In 10 and 20 dimensions, consecutive states are about five clean-state widths apart (Table 4), so the clean density really is a string of separated beads. The noise merges them, which is why histograms and KDEs look continuous. The deconvolved ft-de reproduces the oracle's qualitative pattern (Figure 14): beaded arms, a separate central cluster, and isolated blobs on the lower loops. It does not match individual bead positions. Its 2D relative L^2 is 0.20–0.46 against the discrete

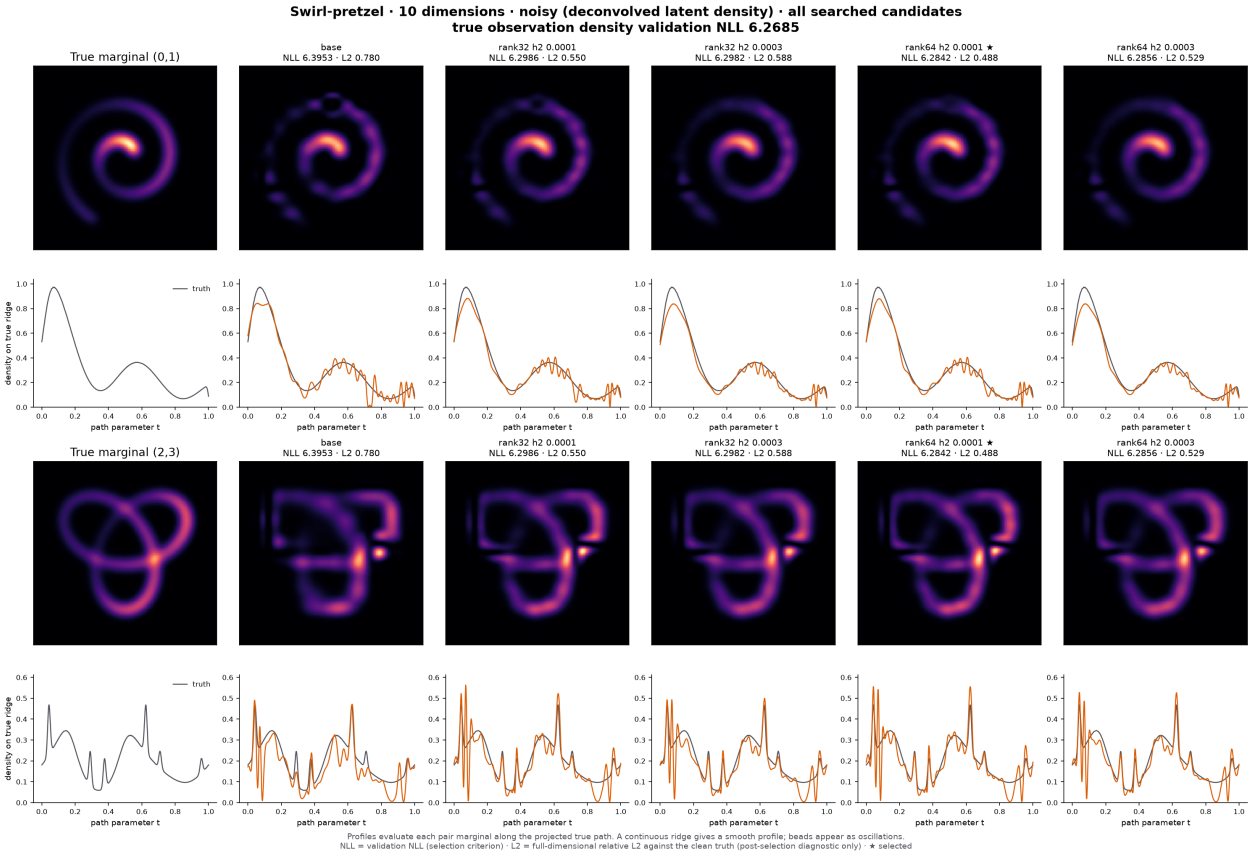


Figure 9: Every searched candidate for the $d = 10$ noisy swirl–pretzel (base, rank 32 and 64, H^2 strengths 10^{-4} and $3 \cdot 10^{-4}$). The profiles underneath evaluate each pair marginal along the projected true path (gray: truth; orange: estimate). The pretzel's spurious blob and straight segments are present in the base fit and in every continuation.

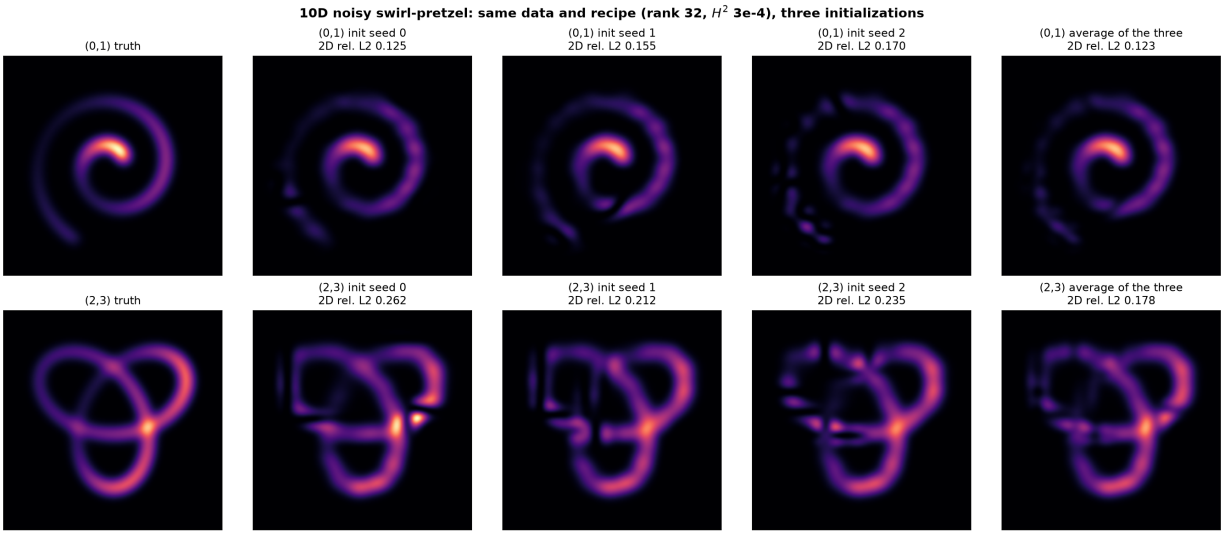


Figure 10: $d = 10$ noisy swirl–pretzel: the same data and recipe from three chebdGE initializations, and the average of their normalized densities. The noisy validation NLL cannot distinguish the four (gaps to the truth 0.030–0.033 $\pm$ 0.004 nats), but the artifacts move between initializations and partly cancel in the average.

embedding	consecutive-state spacing	clean state s.d.	within-state s.d.
4D	1.24	0.65	1.19
10D	2.16	0.43	1.08
20D	2.05	0.41	1.07

Table 4: Cryo-EM label oracle, in units of the supplied noise standard deviation (medians over states and axes).

oracle and 0.16–0.38 against the continuous counterfactual, and pointwise L^2 heavily penalizes misplaced beads. In the separately fitted 4D embedding the states are only about two clean widths apart and the oracle is a lumpy but mostly connected ridge. There the estimate is more beaded than the truth.

The swirl–pretzel experiment gives the other half of the answer. At the same noise-to-width ratio, a continuous ridge is recovered continuously in 4D, but in 10D the fit fragments it through a detectably suboptimal basin. For the 10D and 20D cryo-EM fits, both mechanisms are therefore present. The truth is beaded, and the estimator can also create or move beads. Individual beads should not be read as individual states unless they recur across independent initializations (Figure 10); the noisy likelihood alone cannot make this distinction.

9 Discussion and open problems

- **What is exact.** Normalization, marginals, smoothness functionals and the convolved likelihood are all exact contractions. The only numerical approximation is the Chebyshev representation of the prescribed kernel, which is checked a priori. This removes Monte Carlo deconvolution error entirely, and it makes post-hoc diagnostics exact when the truth is known.

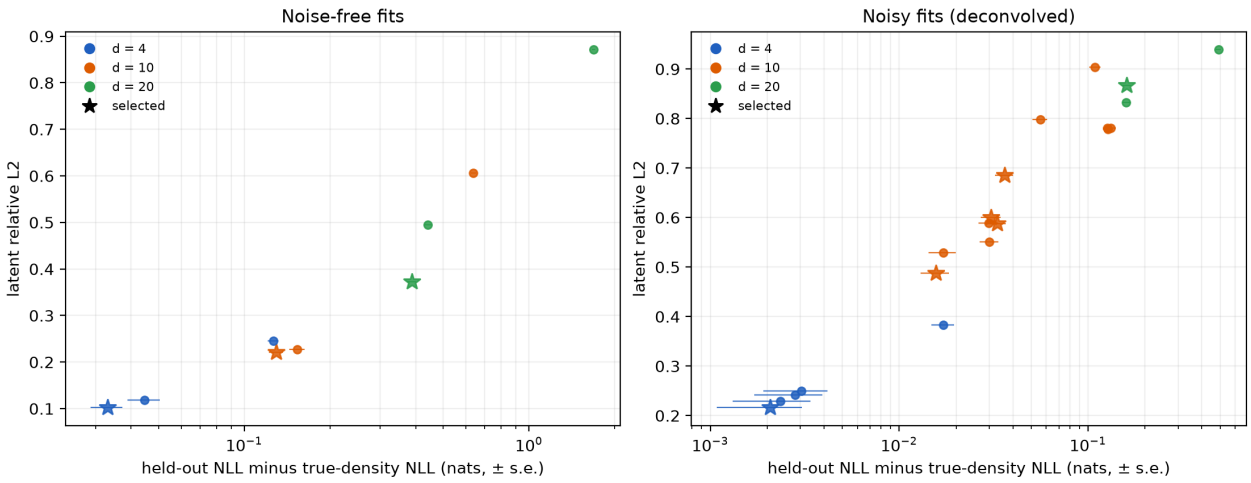


Figure 11: Held-out NLL minus the NLL of the true (observation) density against latent rel L^2, for every searched swirl–pretzel candidate. Stars mark the selected candidates.

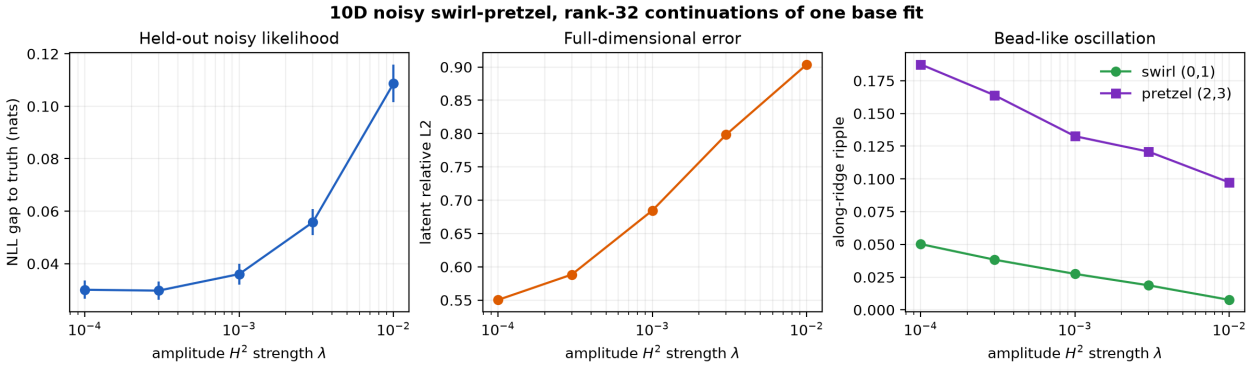


Figure 12: Stronger amplitude smoothing in the $d = 10$ noisy swirl–pretzel (rank-32 continuations of one base fit): the ripple falls, while the latent error and the noisy-likelihood gap rise.

- **Optimization is the bottleneck in higher dimension.** In the swirl–pretzel benchmark the $d = 10$ deconvolution error is associated with a detectable likelihood deficit (0.016–0.03 nats) that neither continuation, stronger smoothing nor re-initialization removes. Different initializations land in equally likely basins whose artifacts differ in location, and averaging their densities cancels part of them without any change in likelihood. Ensembles over initializations, annealed regularization and rank-adaptive growth are the natural next steps; we have not established which of them closes the gap.
- **Selection.** Noisy held-out NLL ordered the candidates correctly here. However, it is flat at the scale of the smallest artifacts in $d = 4$, and it cannot certify a single fit. Stability across independent initializations or resamples is a useful complementary diagnostic.
- **Rates.** For Gaussian noise the minimax rate is logarithmic [6]. Our fixed-capacity estimator does not adapt its rank or degree to N. Over $N = 10^4$–$3 \cdot 10^5$ its deconvolution error falls like $N^{-0.2}$–$N^{-0.25}$, against about $N^{-0.5}$ without noise for GL, and for GL it levels off at the largest N (§8.2). Rank, degree and step budgets that grow with N are needed to tell a statistical floor from a computational one. A theory for squared low-rank models under supersmooth deconvolution is open.

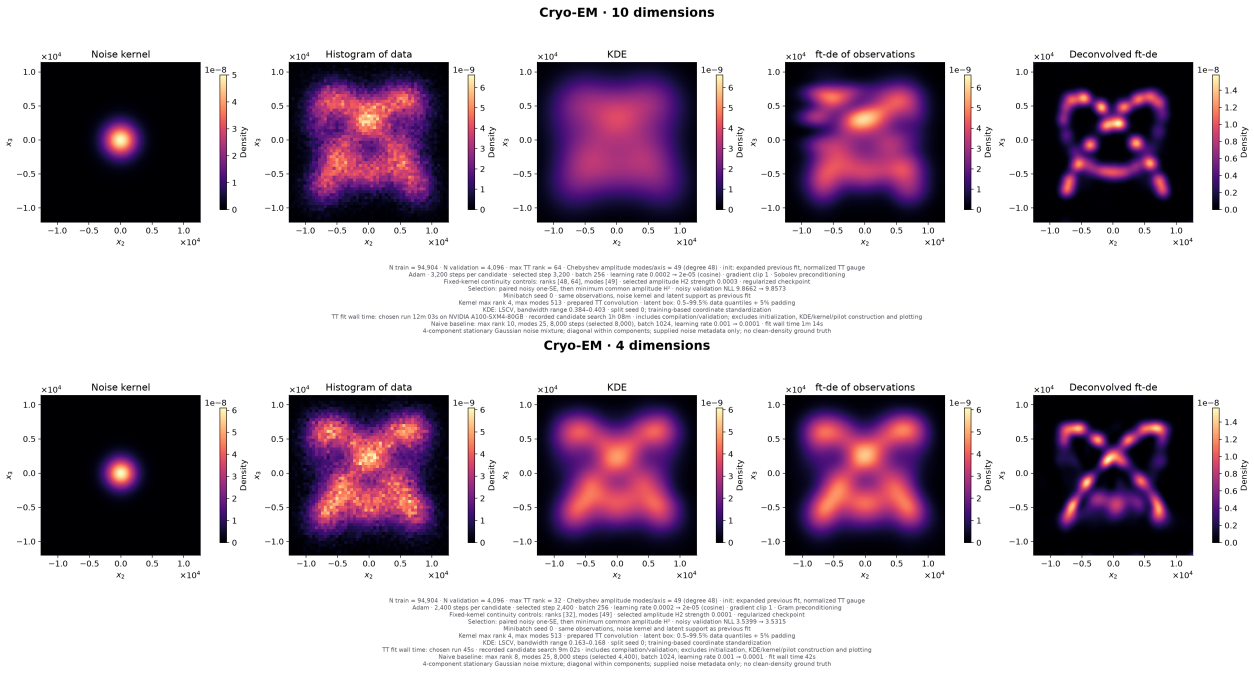


Figure 13: Cryo-EM, coordinates $(2, 3)$ of the 10D (top) and 4D (bottom) embeddings: supplied noise kernel, data histogram, KDE, a TT fitted to the observations (no deconvolution), and the deconvolved ft-de.

- **Cost.** One convolved likelihood evaluation costs $O(d(r^3sn + r^2s^2n^2))$ per observation. Rank 64 with a rank-30 kernel is feasible on one GPU, but memory forced gradient accumulation. The noise-free model is $O(dr^2n)$ per point.

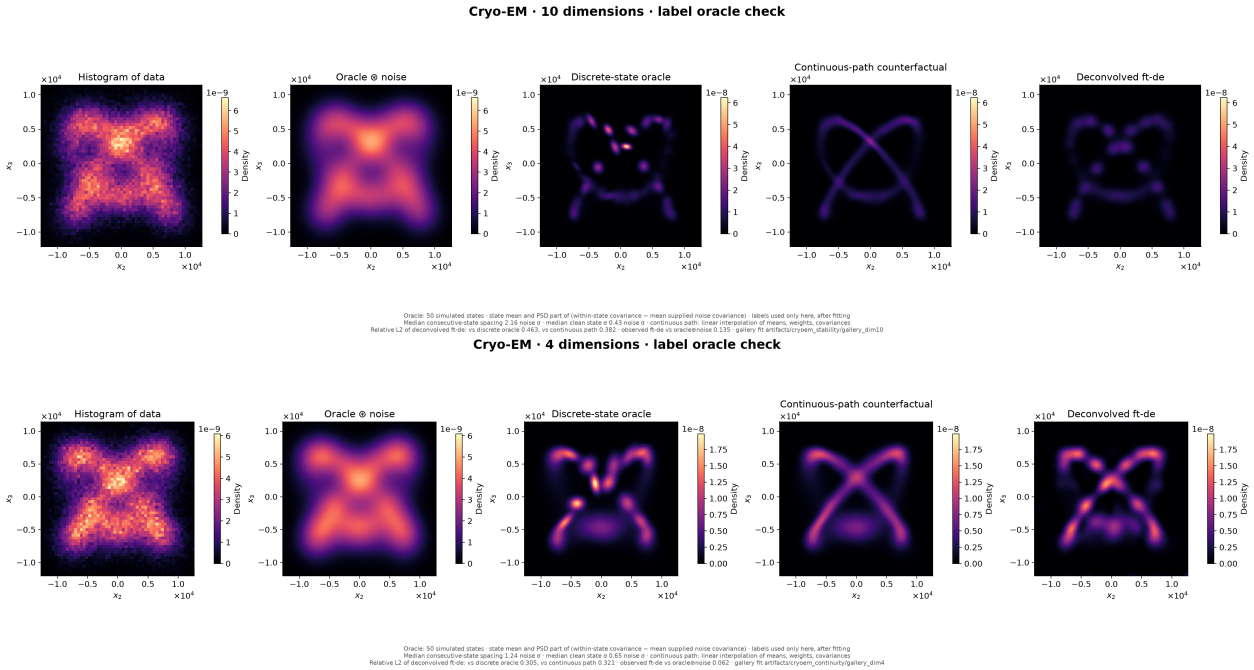


Figure 14: Label oracle for the 10D (top) and 4D (bottom) embeddings: data histogram, oracle $*$ noise, discrete-state oracle, continuous-path counterfactual, deconvolved ft-de. The last three share a color scale.

A Code map and reproduction

concept	module / script
Chebyshev Gram matrices, moments (5)	`ftde/cheb_gram.py`
squared-TT NLL, Z (4)–(8)	`ftde/tt_density.py`
exact pair marginals (7)	`ftde/marginals.py`
KDE / LSCV, cross-approximation init	`ftde/kde.py`, `ftde/init.py` (chebdGE `tt_qtensge`)
noise TT kernels	`ftde/noise_kernel.py`
exact convolved likelihood (Prop. 4)	`ftde/tt_convolution.py`
amplitude Sobolev penalty (12)	`ftde/sobolev.py`
Fisher-visibility filter (15), pilots	`ftde/fisher_visibility.py`, `ftde/marginal_reference.py`
paired one-SE selection	`ftde/noisy_model_selection.py`
Algorithm 1	`scripts/refresh_density_fits.py`
Algorithm 2	`scripts/refine_cryoem_continuity.py`, `scripts/swirl_fits.py`
swirl–pretzel truth	`ftde/swirl.py`; figures `scripts/swirl_figures.py`, `swirl_ripple.py`
cryo-EM label oracle	`scripts/audit_cryoem_oracle.py`
sample-size sweep	`scripts/run_convergence_sweep.py`, `scripts/convergence_summary.py`
figures of this note	`scripts/note_figures.py`

Run commands from the repository root in the `chebdGE` environment, inside one GPU allocation (`salloc`, then `srun` per step), for example `python scripts/swirl_fits.py --dim 10 --noise mixture --output artifacts/swirl/swirl_mixture_dim10`. Completed candidates are reused, and runs refuse to overwrite a directory whose settings differ. See `docs/swirl_benchmark.md` and `docs/figure_refresh.md` for full settings.

References

[1] D. Bigoni, A. P. Engsig-Karup, and Y. M. Marzouk. Spectral tensor-train decomposition. *SIAM Journal on Scientific Computing*, 38(4):A2405–A2439, 2016.

[2] A. W. Bowman. An alternative method of cross-validation for the smoothing of density estimates. *Biometrika*, 71(2):353–360, 1984.

[3] R. J. Carroll and P. Hall. Optimal rates of convergence for deconvolving a density. *Journal of the American Statistical Association*, 83(404):1184–1186, 1988.

[4] T. Cui and S. Dolgov. Deep composition of tensor-trains using squared inverse Rosenblatt transports. *Foundations of Computational Mathematics*, 22:1863–1922, 2022.

[5] S. Dolgov, K. Anaya-Izquierdo, C. Fox, and R. Scheichl. Approximation and sampling of multivariate probability distributions in the tensor train decomposition. *Statistics and Computing*, 30(3):603–625, 2020.

[6] J. Fan. On the optimal rates of convergence for nonparametric deconvolution problems. *The Annals of Statistics*, 19(3):1257–1272, 1991.

[7] M. A. Gilles and A. Singer. Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression. *Proceedings of the National Academy of Sciences*, 122(9): e2419140122, 2025.

[8] A. Gorodetsky, S. Karaman, and Y. Marzouk. A continuous analogue of the tensor-train decomposition. *Computer Methods in Applied Mechanics and Engineering*, 347:59–84, 2019. doi: 10.1016/j.cma.2018.12.015.

[9] B. Hashemi and L. N. Trefethen. Chebfun in three dimensions. *SIAM Journal on Scientific Computing*, 39(5):C341–C363, 2017.

[10] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning*. Springer, 2 edition, 2009.

[11] Y. Hur, J. G. Hoskins, M. Lindsey, E. M. Stoudenmire, and Y. Khoo. Generative modeling via tensor train sketching. *Applied and Computational Harmonic Analysis*, 67:101575, 2023.

[12] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

[13] G. S. Novikov, M. E. Panov, and I. V. Oseledets. Tensor-train density estimation. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*, volume 161 of *Proceedings of Machine Learning Research*, pages 1321–1331, 2021.

[14] I. Oseledets and E. Tyrtyshnikov. TT-cross approximation for multidimensional arrays. *Linear Algebra and its Applications*, 432(1):70–88, 2010.

[15] I. V. Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5): 2295–2317, 2011.

[16] Y. Peng, Y. Khoo, and D. Wang. Multivariate density estimation via variance-reduced sketching, 2024.

[17] M. Rudemo. Empirical choice of histograms and kernel density estimators. *Scandinavian Journal of Statistics*, 9(2):65–78, 1982.

[18] D. W. Scott. *Multivariate Density Estimation: Theory, Practice, and Visualization*. Wiley, 2 edition, 2015.

[19] L. A. Stefanski and R. J. Carroll. Deconvolving kernel density estimators. *Statistics*, 21(2): 169–184, 1990.

[20] A. Townsend and L. N. Trefethen. An extension of Chebfun to two dimensions. *SIAM Journal on Scientific Computing*, 35(6):C495–C518, 2013.

[21] L. N. Trefethen. *Approximation Theory and Approximation Practice, Extended Edition*. SIAM, Philadelphia, 2019.